Zayn Furnishings

Product Requirements Document (PRD) - E-commerce Website

Item	Detail
Version	v0.1 (draft for review)
Date	November 7, 2026
Product	US-only online furniture store with owner admin panel
Prepared by	Kaif (developer)
Status	Project skeleton running locally; web and API connected to Supabase Postgres
Repository	github.com/Mohd16kaif/zayn-furnishings

How to read this document. Items marked **[Decided]** were agreed in planning. Items marked **[Assumed]** are sensible defaults that need client confirmation. Section 17 lists open questions.

1. Overview

Zayn Furnishings is an online furniture store serving customers in the United States only. The client owns a furniture business and wants a website with the look, navigation and shopping flow of large furniture retailers such as Ashley (ashleyfurniture.com), which he selected as the reference. The site must let the client run the whole business himself: list products, manage stock, take orders and payments, message customers by email, SMS and WhatsApp, and arrange deliveries, all without technical knowledge.

1.1 Goals

- Launch an original, professional storefront that feels familiar to shoppers of large furniture retailers.
- Give the owner a simple, in-depth admin panel to manage categories, products, stock, orders, deliveries and customers.
- Take secure online payments and calculate US sales tax correctly.
- Keep customers informed through email, SMS and WhatsApp.
- Run on free tiers during development and early traffic, with a clear upgrade path as the store grows.

1.2 Non-goals (not in the first release)

- Selling outside the US, multi-currency or multi-language.
- Product variants (size, color, fabric). Each color or size is listed as its own product. [Decided]
- Marketplace features, multiple vendors, or a mobile app.
- In-house financing. Third-party financing (Affirm, Klarna) is planned after launch. [Assumed]
- 3D or AR product viewers.

1.3 Important note on the reference site

Ashley is used as a **UX reference only**: layout patterns, navigation structure and page flow. Its source code, images, logo, text and branding are copyrighted and will not be copied. The Zayn Furnishings design, copy, imagery and brand assets will be original or supplied by the client. [Decided]

2. Users and Roles

Role	Description	Key needs
Visitor / Guest	Anyone browsing the site	Browse, search, filter, view products, add to cart, check delivery by ZIP
Customer	Verified by email OTP at first checkout	Place orders, track orders, view order history, save addresses, wishlist
Owner (admin)	The client	Full access: catalog, stock, orders, deliveries, customers, settings, staff accounts
Staff (admin)	Optional employees	Limited access: manage orders and deliveries; no settings or staff management [Assumed]
Developer	Kaif	Deploys, monitors, fixes bugs, supports the client under a support agreement

3. Scope

3.1 MVP (launch scope)

- Storefront: home, category, product, search, cart, checkout, order confirmation, order tracking, account, static and policy pages.
- Passwordless login (email OTP) at first checkout and for viewing order history. [Decided]
- Admin panel at /admin: categories, products, images, stock, orders, deliveries, delivery zones, customers, banners and homepage sections, discount codes, staff.
- Payments with Stripe (cards, Apple Pay, Google Pay); sales tax with Stripe Tax. [Assumed]
- Email notifications; SMS notifications once US carrier registration is approved; WhatsApp click-to-chat button.
- Delivery fee rules by ZIP code; manual delivery scheduling and status updates by the owner.
- Accessibility (WCAG 2.1 AA target), SEO basics, privacy and legal pages.

3.2 Phase 2 (after launch)

- Financing options (Affirm or Klarna through Stripe).
- WhatsApp Cloud API with approved message templates; shared inbox (e.g. Chatwoot) for two-way email, SMS and WhatsApp.
- Reviews and ratings, product comparison, wishlist sharing, recently viewed.
- Live shipping rates (Shippo or EasyPost) for small parcel items.
- Product variants, if the catalog outgrows the one-product-per-option approach.
- Abandoned-cart reminders, email marketing, analytics dashboard in admin.

4. Reference Analysis: Ashley Homepage

The screenshots reviewed show the patterns below. The right-hand column states how Zayn Furnishings will implement each pattern with original content.

Pattern seen on reference	Zayn Furnishings implementation
Dark top bar with nearest store and a 'Delivering to ZIP' selector	Top bar with 'Delivering to [ZIP]' selector. Saved in a cookie and used to show delivery fee and availability on product pages. Store location shown if the client has a showroom.
'Shop by phone' number in the header	Client phone number in header, tap-to-call on mobile.
Header: search box left, logo center, links right (Login, Order Tracking, Financing, Help, Wishlist, Cart icon)	Same layout. Financing link is added in Phase 2. Cart icon shows item count.
Main category navigation: Living Room, Bedroom, Dining, Mattresses, Rugs and Decor, Office, Baby and Kids, Outdoor, Deals, Trends	Category navigation driven by the categories table (admin-editable). Mega-menu on hover with subcategories and featured image. Mobile: slide-in menu.
Thin promo strip (e.g. discounts for veterans, sale event, financing)	Admin-editable announcement strip with up to three messages.
Dark financing banner under the header	Optional admin-controlled banner (used for sales or financing once enabled).
Full-width hero image with headline and two buttons	Hero carousel managed in admin: image, headline, button text and link, start and end dates.
Two-card promo row (hot buys and financing)	Two configurable promo cards below the hero.
Large room/category image tiles with a 'Shop' button	'Shop by room' tiles linked to categories.
Horizontal deal carousels (e.g. Black Friday deals, Mattress savings)	Product or collection carousels. Admin chooses which products or categories appear and their order.

Pattern seen on reference	Zayn Furnishings implementation
'More ways to shop' cards (financing, free shipping, in stock and ready to deliver)	Info cards: delivery information, financing (Phase 2), in-stock items.
'Our values' block and collapsible 'About' accordion with SEO text	About-us section and FAQ accordion; SEO text managed in admin.
Footer: four link columns, newsletter signup with first-purchase discount, phone, social icons, long legal disclaimers	Footer with About, Help, Services, Legal columns; newsletter signup with opt-in; social links; legal text written for Zayn Furnishings and reviewed by the client.

5. Functional Requirements

5.1 Storefront

ID	Requirement	Priority
SF-01	Home page with announcement strip, hero carousel, promo cards, shop-by-room tiles, product carousels, info cards, FAQ, footer, all admin-editable.	Must
SF-02	Mega-menu navigation built from the category tree; mobile slide-in menu.	Must
SF-03	Category page: product grid, subcategory links, sort (price, newest, name), filters (price range, color, material, in stock), pagination.	Must
SF-04	Product page: image gallery with zoom, name, price, sale price, SKU, stock status ('In stock' / 'Sold out'), description, dimensions, material, color, add to cart, delivery estimate by ZIP, related products.	Must
SF-05	Search with suggestions, searching name, SKU, category and description.	Must
SF-06	Cart: add, remove, change quantity, stock check, saved for guests (cookie) and logged-in customers, discount code entry, estimated delivery and tax.	Must
SF-07	'Delivering to ZIP' selector that validates US ZIPs and shows delivery fee or 'not available'.	Must
SF-08	Static pages: About, Contact, FAQ, Delivery information, Returns, Privacy, Terms, Accessibility, SMS terms.	Must
SF-09	Order tracking page: look up an order by email OTP or order number plus email.	Must
SF-10	Newsletter signup with explicit consent.	Should
SF-11	Wishlist (saved items for logged-in customers).	Should
SF-12	Reviews and ratings; product comparison.	Could (Phase 2)

5.2 Customer accounts and checkout

- Guest can browse and fill a cart without an account.
- At checkout the customer enters an email. A one-time code (OTP) is sent; entering it verifies the email and creates the customer record. Supabase Auth is used for OTP. [Decided]
- Returning customers log in with an email OTP to see order history and saved addresses.
- Checkout steps: contact and verification, delivery address, delivery method and date preference, payment, review, confirmation.
- SMS opt-in is a separate, unchecked checkbox with clear consent text; phone is optional.
- Order confirmation page and email with order number, items, totals, tax, delivery details.

5.3 Payments, tax and delivery

- Stripe Checkout or Payment Element for cards and wallets. The site never stores card data (keeps PCI scope minimal).
- Order is created as 'pending' and becomes 'paid' only after a verified Stripe webhook. Orders must never depend on the customer's browser returning to the site.
- Sales tax calculated by Stripe Tax from the delivery address. [Assumed]
- Delivery fee from delivery_zones: flat fee, per-ZIP fee, or free over a configurable order amount. The owner edits these in admin. [Assumed]
- Stock decreases automatically when payment succeeds; out-of-stock products stay visible and show 'Sold out'. [Assumed]
- Refunds and cancellations are triggered by the owner in admin and executed through Stripe.

5.4 Admin panel (/admin)

Built for a non-technical owner: plain labels, confirmation before destructive actions, and inline help text.

Area	Capabilities
Dashboard	Today's orders and revenue, low-stock list, orders needing action, recent activity.
Categories	Create, rename, reorder, nest (parent/child), set image, hide or show.
Products	Create, edit, duplicate, archive or delete; name, slug, description, price, sale price, SKU, stock quantity, dimensions, weight, material, color, category, multiple images with drag-to-reorder, status (draft or active), SEO title and description; bulk price and stock update; CSV import and export.
Stock	Quick stock editor, low-stock threshold, stock history log.
Orders	List with search and filters (status, date, customer); order detail; update status; add internal notes; issue refund; print packing slip and invoice.
Deliveries	Assign delivery method, schedule date and time window, add tracking details or driver notes, mark out for delivery and delivered; triggers customer notification.
Delivery zones	Manage ZIP codes or ranges with fees and free-delivery thresholds.
Customers	List, search, order history, contact details, SMS opt-in status.
Content	Announcement strip, hero banners, promo cards, shop-by-room tiles, featured collections, FAQ and static pages.
Discounts	Percentage or fixed codes with start/end dates, minimum order, usage limits.
Messages log	Every email and SMS sent, with status, so the owner can verify what customers received.
Staff and settings	Add staff users, set roles, store details, tax and notification settings.

5.5 Notifications

Event	Email	SMS	WhatsApp
OTP / login code	Yes	Later (optional)	No
Order placed	Yes	If opted in	Phase 2
Payment failed or refunded	Yes	If opted in	Phase 2
Delivery scheduled or rescheduled	Yes	If opted in	Phase 2

Event	Email	SMS	WhatsApp
Out for delivery / delivered	Yes	If opted in	Phase 2
Marketing messages	Only with consent	Only with consent	Only with consent

Customers can reply to emails. SMS must honor STOP and HELP. Phase 2 adds a shared inbox so the owner can answer email, SMS and WhatsApp conversations in one place.

6. System Architecture

Layer	Technology	Hosting
Storefront and admin UI	Next.js (App Router), TypeScript, Tailwind CSS. Admin lives under /admin in the same app. [Decided]	Vercel
API	Node.js, Express, TypeScript, zod validation, helmet, CORS	Render
Database	PostgreSQL with Drizzle ORM and drizzle-kit migrations	Supabase (East US, Ohio)
Auth	Supabase Auth: email OTP for customers; separate admin_users table with roles	Supabase
File storage	Supabase Storage for product and banner images	Supabase
Payments and tax	Stripe, Stripe Tax	Stripe
Email	Resend (or similar) as transactional provider and Supabase SMTP	Resend
SMS	Twilio (A2P 10DLC registration required)	Twilio
WhatsApp	Click-to-chat button at launch; Meta WhatsApp Cloud API in Phase 2	Meta
Repo and CI	pnpm monorepo: apps/web, apps/api, packages/shared; GitHub main branch	GitHub

6.1 Request flow

Browser -> Next.js (Vercel) -> Express API (Render) -> Supabase Postgres. All data access goes through the API; the browser never queries the database directly. The API holds the Supabase service role key; the web app holds only public values. Stripe and Twilio call the API through webhooks.

6.2 Free-tier constraints

- **Vercel Hobby** is for non-commercial use. A live client store needs a paid plan before launch.
- **Render free web services sleep** when idle, causing slow first requests. Do not take real orders on the free tier; upgrade the API before launch.
- **Supabase free projects pause** after inactivity and have limited storage and email sending. Upgrade before launch; use a custom SMTP provider for OTP emails.
- Use the Supabase **session pooler** connection string (port 5432), not the transaction pooler, for the API.
- Costs for all services are billed to the client's own accounts, with the developer added as a collaborator. [Assumed]

7. Data Model (high level)

All money is stored as integer cents. Every table has id, created_at and updated_at. Row Level Security is enabled on all tables with no public policies, so data is reachable only through the API.

Table	Key fields	Notes
categories	name, slug, parent_id, image_url, sort_order, is_active	Tree for mega-menu
products	name, slug, description, price_cents, sale_price_cents, sku, stock_qty, low_stock_threshold, width/depth/height, weight, material, color, status, category_id, seo fields	No variants
product_images	product_id, url, alt_text, sort_order	Multiple per product
customers	auth_user_id, email, name, phone, sms_opt_in, sms_opt_in_at	Linked to Supabase Auth
addresses	customer_id, name, line1, line2, city, state, zip, phone, is_default	US addresses
carts / cart_items	customer_id or session_id, product_id, quantity	Guest and customer carts
orders	order_number, customer_id, status, subtotal, discount, delivery_fee, tax, total, shipping address snapshot, stripe_payment_intent_id	Status: pending, paid, processing, out for delivery, delivered, cancelled, refunded
order_items	order_id, product_id, name_snapshot, sku_snapshot, unit_price_cents, quantity	Snapshot of price at purchase
payments	order_id, provider, provider_id, amount, status, raw_event_id	Stripe records
deliveries	order_id, method, scheduled_date, time_window, tracking_info, status, notes	Manual or carrier
delivery_zones	zip_start, zip_end, fee_cents, free_over_cents, is_active	Owner-editable
discount_codes	code, type, value, min_order, starts_at, ends_at, usage_limit, used_count	
content_blocks	type (hero, promo, strip, tile), title, image_url, link, sort_order, starts_at, ends_at	Homepage content
admin_users	auth_user_id, email, role (owner, staff), is_active	Role-based access
stock_movements	product_id, change, reason, order_id, admin_user_id	Audit trail
notifications_log	customer_id, order_id, channel, template, status, provider_message_id	Email and SMS log
audit_log	admin_user_id, action, entity, entity_id, details	Who changed what

8. API Outline

Group	Endpoints (examples)	Access
Public catalog	GET /categories, GET /products, GET /products/:slug, GET /search, GET /content/home	Public
Delivery	GET /delivery/check?zip=	Public
Cart	GET/POST/PATCH/DELETE /cart	Guest session or customer
Auth	POST /auth/otp/request, POST /auth/otp/verify	Public, rate-limited

Group	Endpoints (examples)	Access
Checkout	POST /checkout/session, POST /webhooks/stripe	Customer; Stripe signature
Customer	GET /me, GET /me/orders, GET /me/orders/:id, address CRUD	Customer
Admin catalog	CRUD /admin/categories, /admin/products, image upload, CSV import	Admin
Admin operations	/admin/orders, /admin/deliveries, /admin/delivery-zones, /admin/customers, /admin/discounts, /admin/content	Admin by role
Messaging	POST /webhooks/twilio (status, STOP/HELP), internal notification service	Twilio signature; internal

9. Security Requirements

- Never store card data; Stripe handles it. Verify Stripe and Twilio webhook signatures.
- Supabase service role key lives only in the API environment. No secrets in Git; .env files are ignored and only .env.example is committed.
- Admin routes require an authenticated admin_users record with a role check on every request, not only in the UI.
- Rate-limit OTP requests and login attempts; short OTP expiry; generic error messages.
- Validate all input with zod; parameterized queries through Drizzle; escape output; secure cookies; strict CORS limited to the web origin.
- Enable Row Level Security on all Supabase tables with no public policies.
- Validate image uploads (type and size); store in Supabase Storage with controlled access.
- Audit log for admin changes; regular database backups and a tested restore process (paid Supabase plan).
- Dependency updates and security patches as part of the support agreement.

10. US Compliance Requirements

Area	Requirement
Accessibility (ADA / WCAG 2.1 AA)	Keyboard navigation, alt text, color contrast, form labels, focus states, accessibility statement. E-commerce sites are frequent lawsuit targets.
Payments (PCI)	Use Stripe-hosted fields only; no card data on the server.
Privacy (CCPA/CPRA)	Privacy policy, notice at collection, 'Do not sell or share my personal information' mechanism if applicable, cookie notice.
Email (CAN-SPAM)	Physical address and unsubscribe link in marketing emails; honor opt-outs.
SMS (TCPA, carrier rules)	A2P 10DLC brand and campaign registration, explicit opt-in, STOP/HELP handling, SMS terms and privacy language on the site. Registration can take weeks, so it starts early.
Sales tax	Collect and remit tax by state; the client should consult a tax professional. Stripe Tax helps with calculation.
Furniture rules	California Proposition 65 warnings and flammability labeling where applicable to products.
Legal pages	Terms, returns, shipping and delivery policy, warranty. The client should have these reviewed by a lawyer.

11. Non-functional Requirements

- **Performance:** target Lighthouse performance 85+ on mobile; images optimized and lazy loaded; pages cached where safe.
- **Responsive:** mobile-first layout; most furniture shoppers browse on phones.
- **SEO:** server-rendered pages, clean URLs, sitemap, structured data (Product schema), meta tags editable in admin.
- **Reliability:** orders are never lost if the customer closes the tab; webhook handling is idempotent; stock updates are transactional to prevent overselling.
- **Observability:** error logging and uptime monitoring; alerts for failed webhooks and failed notifications.
- **Maintainability:** TypeScript throughout, shared types package, migrations in Git, README and handover documentation.

12. Delivery Plan and Milestones

Estimates assume part-time to full-time work by one developer using AI-assisted coding. They will shift with client response times and data readiness.

Phase	Work	Estimate
0. Foundation (done)	Monorepo, Next.js, Express, Supabase connection, GitHub repo, health check showing db: true	Complete
1. Database	Drizzle schema, migrations, RLS, seed data	1 week
2. Admin core	Admin auth and roles, categories, products with images, stock	2 weeks
3. Storefront	Home, mega-menu, category, product, search, content blocks, responsive design	3 weeks
4. Cart and checkout	Cart, OTP login, Stripe, Stripe Tax, delivery zones, orders, webhooks	2-3 weeks
5. Operations	Admin orders, deliveries, customers, discounts, dashboard	2 weeks
6. Messaging	Transactional email, SMS (after 10DLC approval), WhatsApp button, notification log	1-2 weeks
7. Hardening	Accessibility, SEO, security review, performance, legal pages, testing with real test orders	2 weeks
8. Launch	Paid hosting upgrade, domain and DNS, production keys, owner training, handover	1 week

Overall, expect about 3 to 4 months of calendar time for a polished launch. Starting the US SMS registration and Stripe verification early avoids waiting on third parties near the end.

13. Roles, Accounts and Ongoing Costs

- The client creates and owns accounts for domain, Vercel, Render, Supabase, Stripe, Twilio and the email provider, and adds the developer as a collaborator.
- The client provides business details (legal name, EIN, address, bank account) needed by Stripe, Twilio and tax setup.
- The client supplies the product spreadsheet, photos, logo, brand colors, policies and delivery rules.
- Expected monthly costs after launch: hosting upgrades for Vercel, Render and Supabase, plus Twilio usage, email provider and domain. Stripe charges per transaction. Exact amounts depend on current provider pricing and

traffic and must be checked before quoting the client.

- A monthly support agreement covers bug fixes, updates, backups, monitoring and small changes.

14. Risks and Mitigations

Risk	Impact	Mitigation
Free tiers cause cold starts or paused databases	Slow or failed checkout	Upgrade Render, Supabase and Vercel before taking real orders
US SMS registration delays	SMS not ready at launch	Start 10DLC registration early; launch with email first
Incorrect tax or double charges	Financial loss, compliance issues	Use Stripe Tax; test with Stripe test mode; idempotent webhooks; owner review of tax setup
Overselling stock	Cancelled orders, unhappy customers	Transactional stock decrement at payment; low-stock alerts
Copying reference-site assets	Legal claims against the client	Original design and content only
Messy or late product data	Delayed launch	Provide a CSV template; collect early; support CSV import
Admin too complex for the owner	Owner cannot manage the store	Plain-language UI, validation, help text, training session, handover guide
Single developer dependency	Outages when developer is unavailable	Documentation, monitoring, clean code, client-owned accounts
Accessibility lawsuit	Legal and financial exposure	Build to WCAG 2.1 AA and test with automated and manual checks

15. Acceptance Criteria for Launch

- A new customer can browse, search, add to cart, verify email with OTP, pay with a test card, and receive an order confirmation email.
- A payment confirmed by Stripe always results in a paid order and reduced stock, even if the browser is closed.
- The owner can create a category and a product with images, change its stock and price, and see the change on the storefront within a minute, without developer help.
- The owner can view an order, schedule delivery, update status, and the customer is notified.
- Delivery fee is correct for sample ZIP codes, and unsupported ZIPs are blocked at checkout.
- Sales tax matches expectations for sample addresses in at least three states.
- Admin pages reject unauthenticated and unauthorized users.
- Key pages pass automated accessibility checks and keyboard-only navigation.
- Production runs on paid, non-sleeping hosting with backups enabled and secrets stored only in provider settings.
- The owner has completed a training session and received the handover document.

16. Decisions Log

Decision	Status
Build from scratch instead of Shopify or Medusa	Decided
Next.js frontend on Vercel; Node/Express API on Render; Supabase Postgres, Auth and Storage	Decided
pnpm monorepo with TypeScript and Drizzle ORM	Decided
Admin panel inside apps/web under /admin	Decided
Under 100 products; no variants at launch	Decided
Passwordless customer login by email OTP at first checkout	Decided
Ashley is a UX reference only; original design and assets	Decided
Prices stored as integer cents	Assumed
Stripe Tax for US sales tax	Assumed
Stock decreases on successful payment; sold-out products stay visible	Assumed
Flexible delivery_zones table (flat, per-ZIP, free-over)	Assumed
Financing (Affirm/Klarna) after launch	Assumed
Owner plus optional staff roles	Assumed

17. Open Questions for the Client

- Final business name, logo, brand colors, tone, and showroom address (if any).
- How are items delivered: own trucks, freight carriers, white-glove, pickup, or a mix? Flat, per-ZIP, or weight/size-based fees?
- Which US states or ZIP codes will be served?
- Return and warranty policy; who inspects and handles damaged deliveries?
- Does the client want financing at launch or later?
- Will staff accounts be needed, and what should they be allowed to do?
- Which email address and phone number should customers see and reply to?
- Is WhatsApp a priority at launch or can it follow after email and SMS?
- Approximate product count, categories, and format of existing product data and photos.
- Launch date expectation, including whether a holiday-season deadline exists.

This PRD is a living document. Next step: design the database schema (Section 7) and create the first migrations.